Candidate's Name:..

Signature:..

Random No.						Personal No.		

(Do not write your School/ Centre Name or Number anywhere on this Booklet.)

545/3
CHEMISTRY
Paper 3
(Practical)
Oct. /Nov. 2024
2 hours

UGANDA NATIONAL EXAMINATIONS BOARD

Uganda Certificate of Education

CHEMISTRY

Paper 3
(Practical)

2 hours

INSTRUCTIONS TO CANDIDATES:

This paper consists of **one compulsory** *examination item.*

Answers to this item are to be written in the spaces provided in this booklet. Use **blue** *or* **black** *ink. Any work done in pencil* **except** *drawings and graph will* **not** *be scored.*

All *working* **must** *be clearly shown. Graph paper will be provided.*

Mathematical tables and silent non-programmable scientific calculators may be used.

You are **not** *allowed to use reference books.*

Candidates are advised to carefully read the item, make sure they have all the apparatus and chemicals they may need and then **plan** *appropriately before starting.*

Turn Over

ITEM

Rashid is a student who intends to make a presentation during a science fair on **Alternative Energy Solutions.** He wants to use the reaction of iron with copper(II) sulphate solution to generate heat. However, he does not know the maximum amount of heat generated by the reaction.

Iron reacts with copper(II) Sulphate according to the following equation:

$$CuSO_{4\,(aq)} + Fe_{(s)} \longrightarrow FeSO_{4\,(aq)} + Cu_{(s)}$$

You have been contacted to help Rashid and you are provided with the following:
BA1, which is 30 cm^3 of copper(II) sulphate solution.
X, which is iron.

Task:

As a learner of Chemistry,

(a) design an experiment you will use to determine the maximum heat change for the reaction.
(*Your design should include; aim, hypothesis, variables, apparatus and materials, procedure, risks and their mitigations.*)

(b) carry out the experiment and record your results.
(*A minimum of six readings required.*)

(c) analyse your results and inform Rashid accordingly.

END

UCE

Do not write in this margin

UGANDA NATIONAL EXAMINATIONS BOARD
UGANDA CERTIFICATE OF EDUCATION
OCTOBER – NOVEMBER, 2024

Candidate's Name: ....

Signature: ....

Subject: .... **Paper Code** ..../.........

Random No.						
Personal Number						

Do not write in this margin

FOR SCORERS' USE ONLY

Item _____

Basis code												
Score												

Scorer's initials ....

Item _____

Basis code												
Score												

Scorer's initials ....

Item _____

Basis code												
Score												

Scorer's initials ....

Item _____

Basis code												
Score												

Scorer's initials ....

Item _____

Basis code												
Score												

Scorer's initials ....

1. The description of the reagents and chemicals specified below does **not** necessarily correspond with the description in the question paper.

   Candidates must **not** be informed of the differences.

2. Candidates are **not** allowed to use reference books during the examination.

3. In addition to the fittings and substances ordinarily contained in a chemistry laboratory, each candidate will require:

   1 plastic beaker / cup.

   1 stop clock / stop watch.

   1 pipette (20 cm^3 or 25 cm^3).

   1 thermometer ($^-10 - 110\ ^oC$).

   200 cm^3 of distilled water.

   30 cm^3 of **BA1**.

   2.0 g of **X**.

   BA1 is prepared by dissolving 250 g of **W** in distilled water to make 1 litre of solution.

   X and **W** will be provided by **UNEB**.